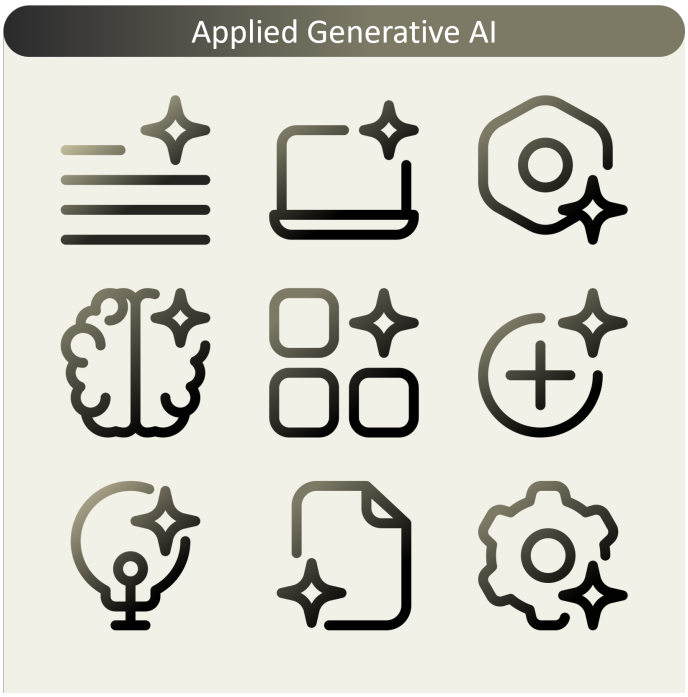
IE 5354: Applied Generative AI



Prof. Mohammad Dehghani

"Prioritize utilizing readily available tools and frameworks instead of developing AI applications from scratch"
- Andrew Ng, Co-founder of Coursera and AI Pioneer

Applied Generative AI

IE 5354:

Instructor Info

Mohammad Dehghani

Office Hrs:

TBD

Website

m.dehghani@northeastern.edu

Course Info

None

Weekly Sessions: As per the itinerary

Time: As mentioned in schedule

TA Info

Check Canvas

Check Canvas

In-class

Check Canvas

Course Overview

Examines how Generative AI (GenAI) and autonomous Agentic AI systems leverage large language models (LLMs) to extract insights, generate solutions, and tackle complex, multi-faceted problems. Emphasizes hands-on exploration of AI-driven approaches and application development across diverse domains, including healthcare, supply chain optimization, financial decision-making, and strategic planning. Provides opportunities to design, deploy, and apply GenAI-powered analytical tools and applications alongside goal-oriented Agentic AI systems. Addresses prompt engineering, model fine-tuning, Retrieval-Augmented Generation (RAG), GenAI ethics, and emerging industry trends.

Note: The course is application-focused and does not deeply cover the fundamental theories of LLMs. Instead, it emphasizes hands-on learning and practical implementation of GenAI.

Note: There are no prerequisites for this course, although familiarity with coding in Python is a plus.

Course Objectives

- Explore the fundamentals of Generative AI and its transformative impact on engineering workflows across multiple domains.
- Acquire hands-on experience in addressing real-world problems through the use of GenAI-enabled tools and the deployment of applications across various domains.
- Develop proficiency in essential concepts such as prompt engineering, Retrieval-Augmented Generation (RAG), and other key techniques that enhance AI capabilities.
- Apply a progressive approach from preliminary GenAI-assisted models to Agentic AI, thereby fostering adaptability in tackling advanced challenges.
- Cultivate familiarity with essential tools and platforms such as *LangChain, LangGraph, n8n* for AI workflow automation, and *Streamlit, Gradio* for interactive application development.

Reference Books

Research papers and open-source tools will provide insights into GenAI applications in engineering.

Outline

Highlights of some of the course's tentative topics and activities:

- Ballots and Bitcoin: Apply GenAI for data analytics in a case study exploring the relationship between presidential elections and Bitcoin prices.
- Supply Chain with Beer Game: Apply GenAI to predict demand and optimize decisions in the Beer Game simulation.
- Decision Support: Demonstrate how to develop a decision-making model enabled by virtual experts created using GenAI.
- MDP for Health Stages: Leverage GenAI to model health transitions with Markov Decision Processes.
- AI-enabled Optimization: Leverage GenAI to address classical engineering problems such as scheduling, resource planning, etc.
- GenAI-Enhanced Excel Analytics: Use GenAI with Excel for advanced analytics like trend and anomaly detection.
- GenBI: Employ GenAI to enhance Business Intelligence (BI) models with insightful visualizations.
- Fine-Tuned LLM Applications: Explore fine-tuning LLMs for custom applications like domain-specific modeling.
- General Applicability of GenAI: Discuss how GenAI impacts finance, pharma, clinical trials, and other domains.

Emailing Us?

For efficient responses, please use the following format for email subject lines:

"Course ID - CourseName # - Subject of Email"

Ex: "IE 5354: - Applied GenAI: 'short description of your query'"

Software Requirements

- Course Platforms: Although the course primarily focuses on applications, familiarity with Python basics is essential to complete the programming tasks effectively.
- Software Access: Students should have a COE Virtual Lab (VLab) account to access required software packages. To set up your account, visit CEO VLAB page (Link).
- OpenAI API: Since most case studies use OpenAI's API, students are responsible for setting up their own API keys using https://platform.openai.com/ due to confidentiality purposes. This ensures that each student maintains control over their usage and associated costs, while also adhering to data privacy and security requirements as outlined by OpenAI's policies.
- Additional API Platforms: Although OpenAI will be the main LLM platform, students are also encouraged to gain API access to other platforms such as Google Gemini and Hugging Face (a leading hub for open-source models and tools) to broaden their experience.
- Google Colab: Almost all course labs are conducted in Google Colab, providing a cloud-based environment for coding and experiments. Students only need a Gmail account to access and participate in lab activities.

Class Schedule

Please check CANVAS for the detailed schedule of the course and deadlines. Refer to Figure 1 for a tentative timeline highlighting key milestones.

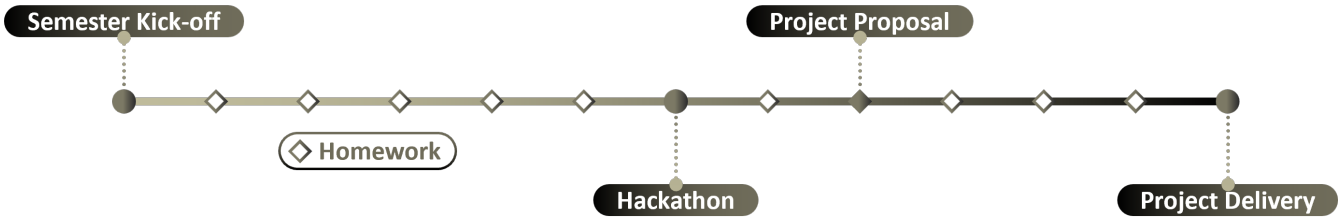
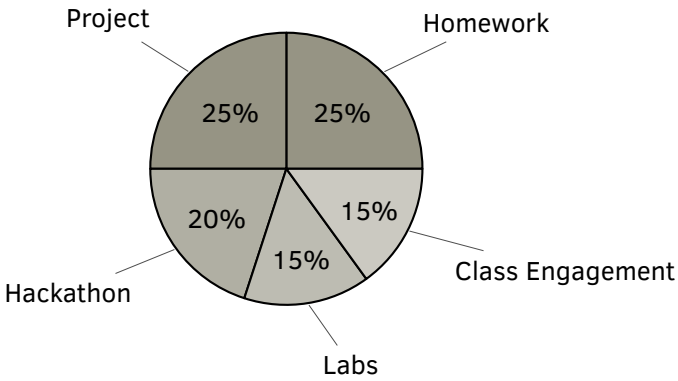


Figure 1: *Tentative course timeline to impart key milestones and deadlines.*

Grading Scheme



- Homeworks: Focus on improving the technical skills of the students. The homework assignments will help students enhance GenAI performance and apply models effectively.

- Labs: The course is fully hands-on, complemented by multiple labs. Students must follow lectures and complete hands-on tasks in labs to reinforce concepts and gain practical experience in GenAI applications. Labs will focus on model fine-tuning, prompt engineering, Retrieval-Augmented Generation (RAG), and deploying AI solutions.

Project: Students will complete the project in three phases: (1) propose a solution to a relevant engineering problem using GenAI models, (2) implement the proposed solution with data and model integration, and (3) present their findings and results.

Hackathon: Students will participate in a timed hackathon to develop and deploy a GenAI-based solution. This event fosters teamwork, creativity, and practical problem-solving skills. Teams will present their solutions, and judging criteria will include innovation, functionality, and presentation quality.

Class Engagement: This category evaluates active participation in class activities, including attendance, in-class and office-hour interactions, as well as team presentations on GenAI topics.

Important Notes:

- Projects, hackathons, and presentations will be completed in teams, while homework assignments are individual.
- Your grades will be posted on CANVAS. You can check your running grade at any time, so there are no surprises. No grade corrections will be made after the last week of the semester.
- This course requires teamwork, and effective communication with teammates is essential for success. We encourage you to read this article which highlights key teamwork skills.

Final grades and course letter grades are not curved and are assigned as follows:

Letter Grade	Points	Letter Grade	Points	Letter Grade	Points
A	93-100	B	83-86	C	73-76
A-	90-92	B-	80-82	C-	70-72
B+	87-89	C+	77-79	F	<70

Class Policy

- No makeup opportunities will be provided for homework, assignments, or projects.
- Students unable to attend classes are responsible for obtaining information regarding missed content.
- Students are expected to attend all class sessions and are responsible for all materials presented.
- Laptops: You may use your laptop during class for course-related activities.
- Phones/tablets/smart watches: Electronic devices must be turned off during class unless required for course work.
- Discussions between students for homeworks and projects are allowed, but individual submissions must reflect individual effort.

Academic Integrity

A commitment to the principles of academic integrity is essential to the mission of Northeastern University. The promotion of independent and original scholarship ensures that students derive the most from their educational experience and their pursuit of knowledge. Academic dishonesty violates the most fundamental values of an intellectual community and undermines the achievements of the entire University.

As members of the academic community, students must become familiar with their rights and responsibilities. In each course, they are responsible for knowing the requirements and restrictions regarding research and writing, examinations of whatever kind, collaborative work, the use of study aids, the appropriateness of assistance, and other issues. Students are responsible for learning the conventions of documentation and acknowledgment of sources in their fields. Northeastern University expects students to complete all examinations, tests, papers, creative projects, and assignments of any kind according to the highest ethical standards, as set forth either explicitly or implicitly in this Code or by the direction of instructors.

Go to Academic Integrity website to access the full academic integrity policy.

Student accommodations

Northeastern University and the Disability Resource Center (DRC) are committed to providing disability services that enable students who qualify under Section 504 of the Rehabilitation Act and the Americans with Disabilities Act Amendments Act (ADAAA) to participate fully in the activities of the university. To receive accommodations through the DRC, students must provide appropriate documentation that demonstrates a current substantially limiting disability.

For more information, visit Disability Resource Center website

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Northeastern University is committed to equal opportunity, affirmative action, diversity and social justice while building a climate of inclusion on and beyond campus. In the classroom, members of the University community work to cultivate an inclusive environment that denounces discrimination through innovation, collaboration and an awareness of global perspectives on social justice. It is my intention that students from all backgrounds and perspectives will be well served by this course, and that the diversity that students bring to this class will be viewed as an asset. I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, socioeconomic background, family education level, ability - and other visible and nonvisible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. Your suggestions are encouraged and appreciated.

Please visit the website for complete information on Diversity and Inclusion.

TITLE IX

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.
Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty and staff.
In case of an emergency, please call 911.

Please visit the website for a complete list of reporting options and resources both on- and off-campus